

AI Prompt Collection

Smart AI-Based E-Learning Web Platform

Using PHP, MySQL, AJAX, JavaScript & Artificial Intelligence

Technology Stack	PHP, MySQL, AJAX, JavaScript, Bootstrap / Tailwind
Architecture	MVC-like, Modular, REST-style endpoints
Sections	20 Development Parts + Future Features
Focus	Modern UI/UX, Security, Scalability, AI Integration

A complete prompt blueprint for designing, building and presenting a professional graduation-level e-learning platform.

Important Development Rules

These rules must be applied across every prompt and every module of the project to keep code quality, security and design consistent.

- Use clean and modern UI/UX.
- Use responsive design.
- Use AJAX for dynamic updates.
- Use secure PHP coding.
- Use MySQL database.
- Use prepared statements.
- Use modular architecture.
- Use reusable components.
- Add comments in code.
- Use Bootstrap or Tailwind CSS.
- Use JavaScript animations.
- Prevent SQL Injection and XSS.
- Use organized folders.
- Use REST-like structure.
- Use professional dashboard design.
- Create mobile-friendly pages.

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Part 1 — Project Structure & Modernization

Prompt

Create a modern professional folder structure for a Smart AI-Based E-Learning Web Platform using PHP, MySQL, AJAX, JavaScript, Bootstrap/Tailwind, and MVC-like architecture.

Requirements

- Separate admin and user panels.
- Create reusable components.
- Create config and database folders.
- Add assets folder.
- Add AJAX folder.
- Add API endpoints folder.
- Add uploads folder.
- Add authentication system structure.
- Add security middleware.
- Create responsive dashboard layouts.
- Add dark mode support.
- Add multilingual support structure.
- Create clean routing system.
- Explain all folders.

Part 2 — Database Design

Prompt

Create a professional MySQL database design for a Smart E-Learning Platform.

Requirements

Tables

- users
- admins
- instructors
- courses
- lessons
- playlists
- categories
- comments
- replies
- likes
- bookmarks
- quizzes
- questions
- answers
- exam_results
- certificates
- notifications
- chats
- messages
- reports
- logs
- user_progress
- enrollments
- course_reviews

General Requirements

- Use proper relationships.
- Add foreign keys.

- Add timestamps.
- Add indexes.
- Add security considerations.
- Normalize database.
- Explain table relationships.
- Add ERD explanation.

Part 3 — Authentication System

Prompt

Create a secure authentication system for a Smart E-Learning Website using PHP, MySQL, AJAX, and JavaScript.

Requirements

Features

- User registration
- Login system
- Logout system
- Remember me
- Forgot password
- Reset password
- Email verification
- OTP verification
- Session management
- Role-based authentication
- Google login integration structure

Security

- Password hashing
- Prepared statements
- CSRF protection
- XSS prevention
- Session security
- Validation

Frontend

- Modern responsive forms
- AJAX validation
- Real-time error messages
- Loading animations

Part 4 — Admin Dashboard

Prompt

Create a modern admin dashboard for the Smart E-Learning Platform.

Requirements

Dashboard Features

- Total users
- Total courses
- Total instructors
- Revenue statistics
- Recent activities
- Notifications
- Course analytics
- Student analytics
- Charts and graphs
- Login logs
- Reports management

Design

- Modern UI
- Responsive design
- Sidebar navigation
- Dark mode
- Interactive charts
- AJAX dynamic updates

Technology

- PHP
- AJAX
- Chart.js
- Bootstrap / Tailwind

Part 5 — Course Management System

Prompt

Create a professional course management system.

Requirements

Features

- Add course
- Edit course
- Delete course
- Upload thumbnails
- Upload videos
- Add PDFs
- Add downloadable files
- Course categories
- Course levels
- Tags system
- Publish/unpublish courses
- Course approval system
- Search and filters

Frontend

- Drag and drop upload
- AJAX upload progress
- Modern cards design
- Dynamic filtering

Security

- Secure file upload
- File validation
- Video protection

Part 6 — Lesson & Video System

Prompt

Create a modern lesson and video streaming system for the e-learning platform.

Requirements

Features

- Video player
- Lesson playlists
- Continue watching
- Watch progress tracking
- Lesson notes
- Fullscreen support
- Playback speed
- Subtitle support
- Auto next lesson
- Download attachments

AJAX Features

- Save progress without refresh
- Dynamic comments
- Real-time updates

UI

- Netflix / Udemy-like design
- Responsive video layout

Part 7 — Quiz & Exam System

Prompt

Create a professional online quiz and examination system.

Requirements

Features

- Multiple choice questions
- True / false questions
- Random questions
- Quiz timer
- Auto grading
- Instant results
- Student ranking
- Exam history
- Passing score system
- Certificate eligibility

AJAX

- Submit answers without refresh
- Real-time timer
- Dynamic question loading

Security

- Prevent cheating
- Timer protection
- Session validation

Part 8 — Certificate System

Prompt

Create an automatic certificate generation system.

Requirements

Features

- Generate PDF certificate
- Add student name
- Add course name
- Add QR verification
- Add unique certificate ID
- Download certificate
- Verify certificate page

Design

- Professional modern certificate design
- Responsive verification page

Part 9 — AJAX Features

Prompt

Upgrade the entire website using AJAX for modern dynamic interactions.

Requirements

Features

- Live search
- Like without refresh
- Bookmark without refresh
- Dynamic comments
- Infinite scrolling
- Notifications updates
- Real-time dashboard updates
- Dynamic filtering
- Load more system
- Live validation

Implementation Requirements

- Use fetch API or jQuery AJAX
- Show loading animations
- Show success / error messages
- Prevent page refresh

Part 10 — Real-Time Chat System

Prompt

Create a real-time chat system between students and instructors.

Requirements

Features

- Private chat
- Real-time messaging
- Online / offline status
- Typing indicator
- Message notifications
- File sharing
- Chat history

Frontend

- Messenger-like UI
- Responsive design
- AJAX real-time updates

Part 11 — Notification System

Prompt

Create a modern notification system.

Requirements

Features

- Real-time notifications
- Course updates
- Exam reminders
- New messages
- Admin alerts
- Notification dropdown
- Mark as read

Technology

- AJAX polling
- Firebase (optional)

Part 12 — AI Chatbot System

Prompt

Create an AI chatbot assistant for the Smart E-Learning Platform.

Requirements

Features

- Answer student questions
- Explain lessons
- Recommend courses
- Provide learning help
- AI-based suggestions

Frontend

- Floating chatbot UI
- Modern animations
- Typing effect

Backend

- PHP API integration structure
- OpenAI API structure

Part 13 — Recommendation System

Prompt

Create an intelligent recommendation system.

Requirements

Features

- **Recommend courses based on:**
 - user interests
 - watch history
 - quiz scores
 - completed courses
- Personalized dashboard
- Trending courses
- Popular courses

Logic

- Recommendation algorithm
- Analytics tracking

Part 14 — Analytics & Reports

Prompt

Create a complete analytics and reports system.

Requirements

Features

- Student progress analytics
- Course popularity analytics
- Revenue analytics
- User activity tracking
- Performance charts
- Monthly reports
- Export reports

Technology

- Chart.js
- AJAX updates
- Admin dashboard integration

Part 15 — Security System

Prompt

Upgrade the security of the e-learning platform.

Requirements

Features

- CSRF protection
- XSS protection
- SQL injection prevention
- Secure sessions
- Login attempt limiter
- File upload validation
- Activity logging
- Admin access control

Documentation

- Security architecture
- Best practices
- Middleware structure

Part 16 — Modern UI/UX Redesign

Prompt

Redesign the entire e-learning platform with modern UI/UX.

Requirements

Design Style

- Modern dashboard
- Clean typography
- Responsive design
- Smooth animations
- Dark mode
- Glassmorphism effects
- Professional color palette

Pages

- Home page
- Courses page
- Course details
- Video lesson page
- Dashboard
- Admin panel
- Authentication pages

Technology

- Bootstrap or Tailwind
- JavaScript animations
- Responsive layouts

Part 17 — Multi-Language System

Prompt

Create a multilingual system for the platform.

Requirements

Languages

- English
- Kurdish
- Arabic

Features

- Language switcher
- Dynamic translation system
- RTL support
- Save selected language

Part 18 — Search & Filter System

Prompt

Create an advanced search and filtering system.

Requirements

Features

- Live search
- Course filters
- Category filters
- Price filters
- Instructor filters
- Search suggestions
- AJAX search results

Part 19 — Responsive Optimization

Prompt

Optimize the entire website for all devices.

Requirements

Devices

- Mobile
- Tablet
- Desktop

Optimization

- Responsive layouts
- Fast loading
- Lazy loading
- Optimized images
- Better UX

Part 20 — Final Project Optimization

Prompt

Perform final optimization and professional improvements for the Smart AI-Based E-Learning Platform.

Requirements

- Optimize performance
- Clean code structure
- Remove duplicate code
- Improve security
- Improve SEO
- Improve accessibility
- Add loading animations
- Improve UX
- Improve database performance
- Create professional documentation
- Create deployment guide
- Create presentation notes
- Create system architecture diagram
- Create API documentation

Future Features

These features can be mentioned in your graduation presentation as future improvements and scalability ideas:

- Mobile App
- AI Voice Assistant
- Blockchain Certificates
- Video Conference Classes
- AI Exam Generator
- Face Recognition Attendance
- AI Course Creator
- Smart Learning Prediction
- Cloud Deployment
- Microservices Architecture

Final Professional Title

Smart AI-Based E-Learning Web Platform

Using PHP, MySQL, AJAX, and Artificial Intelligence